

BRAC UNIVERSITY
Department of Computer Science and Engineering

Set: A

Examination: Final
Duration: 1 Hour 50 Minutes

Semester: Spring 2025
Full Marks: 40

CSE 340: Computer Architecture

Answer the following questions. Show calculations where required. Figures in the right margin indicate marks. Understanding the question is part of the examination.

Name: MD RAKIBUL HASAN	ID: 22101326	Section: 09
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- ✓ **CO1** a) In 3 different arbitrary length IEEE 754 floating point representations, consider the exponent field sizes are respectively 6, 9, and 13 bits. If the biased exponents are respectively $(20)_{10}$, $(94)_{10}$, and $(111)_{10}$, then calculate the actual exponents of the given biased exponents. Your answer should show the detailed calculations. 1.5*
3 =
4.5
✓
- b) Consider $X = (9.321)_{10}$ and $Y = (3.325)_{10}$; Perform $X*Y$ using IEEE754 single precision floating point representation. Your answer should show all the detailed steps. Note: Consider 4 digits after the floating point while converting from decimal to binary. 2.5
✓
- c) In an IEEE754 64-bit format, the exponent field is 7 bits. What is the biased exponent and the actual exponent range? 1+1
= 2
✓

2. **CO2** a) Consider the given RISC-V instructions below. Consider that X_3 and X_4 initially have values $(10)_{10}$ and $(40)_{10}$. 2*3
= 6

Instruction No.	Instruction	Answer the following questions: i. For each of the following instructions 1, 3, 8, and 9, write the corresponding outputs. ii. Consider the memory address of instruction 1 is $(00001004)_{16}$. Calculate the offset that is required for going to instruction 6 from instruction 3 in case of a branch, also calculate the memory address of instruction 9. Your answers should be in Hex. iii. If instruction 10 is a branch instruction, and in case of a branch it jumps to instruction 3, calculate the branch offset. Your answer should be in decimal.
1	Slli $X_3, X_3, 2$	
2	Instruction	
3	Beq $X_3, X_4, Else$	
4	Instruction	
5	Instruction	
6	Else: Instruction	
7	Instruction	
8	Srli $X_4, X_3, 2$	
9	Or X_4, X_4, X_3	
10	Instruction	

- b) What role does the offset play in base displacement addressing in RISC-V? 2
✓
- c) What are immediate operands in RISC-V instructions, and how do they differ from other types of operands? 1+1
= 2
✓

3.
CO3

- a) Consider the following instructions. Identify the components corresponding to the stages in their lifecycle, including the control unit.
(i) Sub, (ii) Sd, (iii) Bne

- b) Design a single-cycle Datapath for the following instruction. Your diagram should have labeling of bit lengths and data flow.

SD $X_1, 1024(X_1)$

- c) Consider the following instruction mix:

R type	I-type (non-load)	Load	Store	Branch
25%	25%	25%	4%	21%

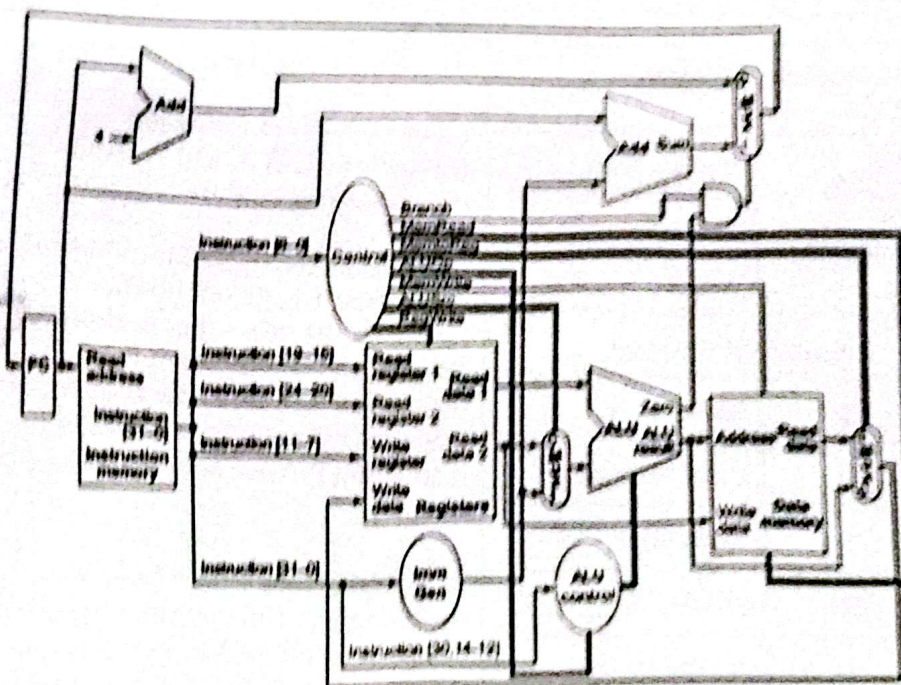
- i) For what fraction of all instructions, the memory read/write control signal is high?
ii) For what fraction of all instructions, only one input from the register file is required?
iii) For what fraction of all instructions, the ALU get input from the immediate word?

- d) In the table below, we examine how pipelining affects the clock cycle time of the processor. This problem assumes that individual stages of the datapath have the following latencies:

IF	ID	EX	MEM	WB
250ps	350ps	150ps	350ps	200ps

- i) What is the clock cycle time in a pipelined and non-pipelined processor?
ii) What is the total latency of an SD instruction in a pipelined and non-pipelined processor?
iii) What is the total latency of an Addi instruction in a pipelined and non-pipelined processor

e)



Consider the above diagram. If you are executing the instruction **Bne X_1, X_0 , Else**, what are the values of control signals generated by the control unit for this instruction? Answer in the table below. For any unused control signal, write 'X'

RegWrite	MemRead	MemWrite	ALUSrc	Branch	MemtoReg
0	0	0	0	1	X

4.
CO4

a) Considering the concept of locality, give two examples each for each type of locality concept. $1 \times 2 = 2$

b) Assume that a memory access takes 10 ns if it results in a cache hit, and 100 ns if it results in a cache miss. The current state of the direct-mapped cache is shown below: $0.5 \times 8 + 1 = 5$

Index	V	Tag	Data
000	N Y	000 10	000 10000
001	N		
010	N Y	000 11 000 10	000 11010 000 10010
011	N Y	00000	000000 11
100	N		
101	N		
110	N Y	000 10	000 10 110
111	N		

The size of a block is 8 bits. Using the table below, calculate the total access time for the given memory sequence using a direct-mapped cache. Indicate hit/miss and access time for each entry. Do the necessary calculations in your script and write your final answers in the table below.

Memory/Address	Hit/Miss	Access Time
22	miss	100 ns
26	miss	100 ns
22	hit	10 ns
16	miss	100 ns
3	miss	100 ns
16	hit	10 ns
18	miss	100 ns
26	hit	10 ns
Total Access Time		530 ns

CSE 340: Computer Architecture

Answer the following questions. Show calculations where required. Figures in the right margin indicate marks. Understanding the question is part of the examination.

Name:

ID:

Section:

1. CO1

- a) In 3 different arbitrary length IEEE 754 floating point representations, consider the 1.5* exponent field sizes are respectively 6, 9, and 13 bits. If the biased exponents are 4.5 respectively $(20)_{10}$, $(94)_{10}$ and $(111)_{10}$, then calculate the actual exponents of the given biased exponents. Your answer should show the detailed calculations. 7.5
- b) Consider $X = (9.321)_{10}$ and $Y = (3.325)_{10}$. Perform $X*Y$ using IEEE754 single precision floating point representation. Your answer should show all the detailed steps. Note: Consider 4 digits after the floating point while converting from decimal to binary. 7.5
- c) In an IEEE754 64-bit format, the exponent field is 7 bits. What is the biased exponent and the actual exponent range? 1+1 = 2

2. CO2 a) Consider the given RISC-V instructions below. Consider that X_3 and X_4 initially have 2*3 values $(10)_{10}$ and $(40)_{10}$. = 6

Instruction No.	Instruction	Answer the following questions:
1	Slli $X_3, X_3, 2$	i. For each of the following instructions 1, 2, 8, and 9, write the corresponding outputs. ii. Consider the memory address of instruction 1 is $(00001004)_{10}$. Calculate the offset that is required for going to instruction 6 from instruction 3 in case of a branch, also calculate the memory address of instruction 9. Your answers should be in Hex. iii. If instruction 10 is a branch instruction, and in case of a branch it jumps to instruction 3, calculate the branch offset. Your answer should be in decimal.
2	Instruction	
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8	Srli $X_4, X_3, 2$	
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- c) What are immediate operands in RISC-V instructions, and how do they differ from other types of operands? 1+1 = 2

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CO3

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(i) Sub, (ii) Sd, (iii) Bne

1+1
+1
= 3

- b) Design a single-cycle Datapath for the following instruction. Your diagram should have labeling of bit lengths and data flow.

2

SD $X_3, 1024(X_5)$

- c) Consider the following instruction mix:

R-type	I-type (non-load)	Load	Store	Branch
25%	25%	25%	4%	21%

1+1
+1
= 3

- i) For what fraction of all instructions, the memory read/write control signal is high?
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iii) For what fraction of all instructions, the ALU get input from the ImmGen unit?

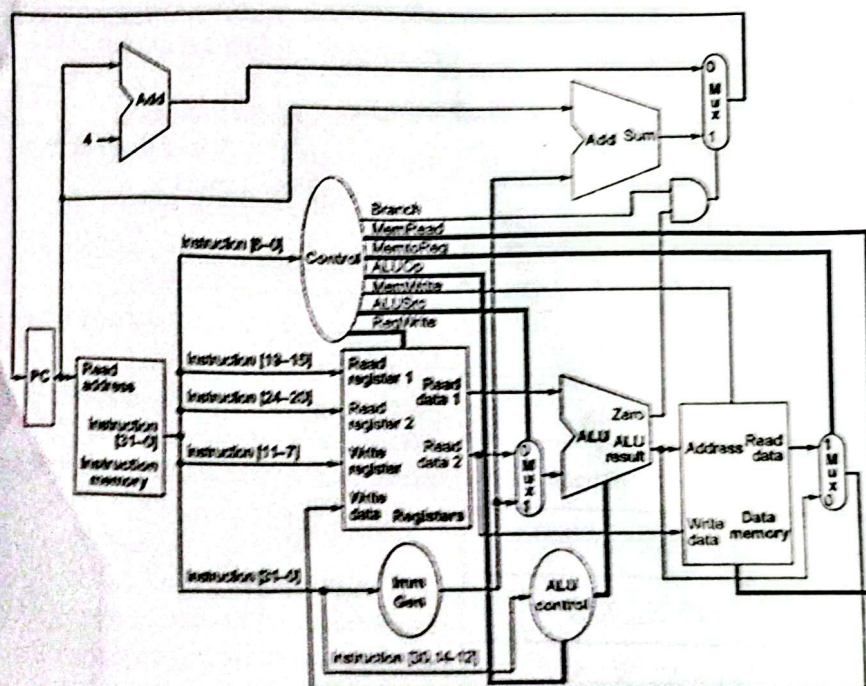
- d) In the table below, we examine how pipelining affects the clock cycle time of the processor. This problem assumes that individual stages of the datapath have the following latencies:

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22		
26		
22		
16		
3		
16		
18		
26		
Total Access Time		